# Practical-2

A. Defination:

Design and Implementation of Half adder, full adder and half substractor, Full substractor circuits

# Half adder:

Truth table

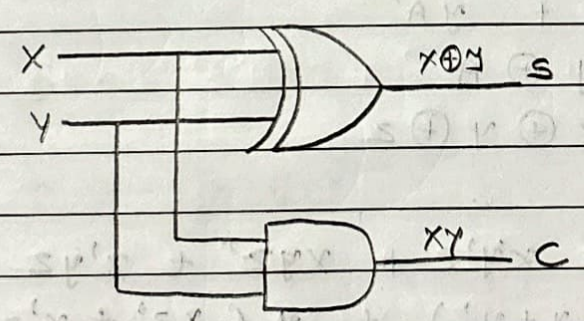
| X | Y | S | C |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 1 |
| 1 | 0 | 1 | 0 |

boolean expression

$S = xy' + yx' = x \oplus y$

$C = xy$

circuit



# Full adder

Truth table

| x | y | z | S | C |
|---|---|---|---|---|
| 1 | 1 | 1 | 1 | 1 |
| 1 | 0 | 1 | 0 | 1 |
| 1 | 1 | 0 | 0 | 1 |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 0 | 1 | 1 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 0 | 1 | 0 | 1 | 0 |
| 0 | 1 | 1 | 0 | 1 |

Expression :

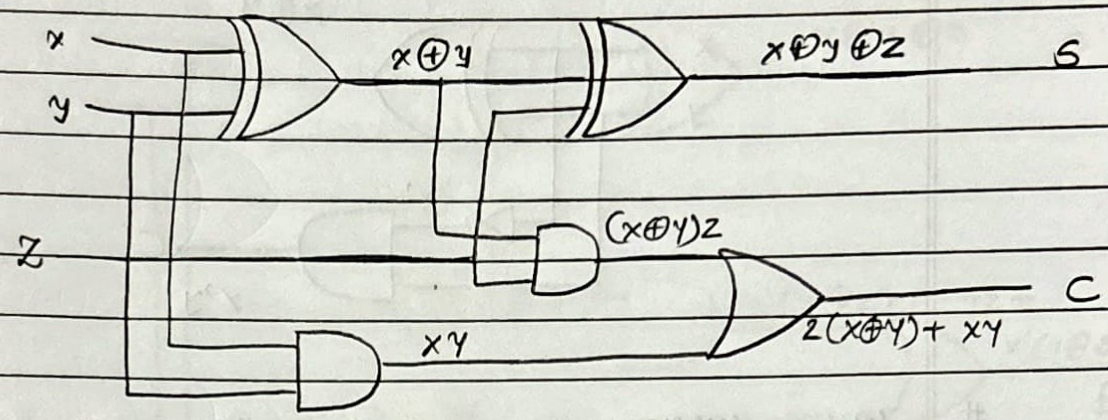
$$S = x'y'z + xy'z' + x'yz' + xyz$$
$$= y'(x'z + xz') + y(x'z' + xz)$$
$$= y'(x \oplus z) + y(x \oplus z)'$$

here assume $x \oplus z = A$

$$= y'A + yA'$$
$$= y \oplus A$$
$$= x \oplus y \oplus z$$

$$C = xyz + xy'z + xyz' + x'yz$$
$$= xz(y + y') + y(xz' + x'z)$$
$$= xz + y(x \oplus z)$$
$$= xy + z(x \oplus y)$$ # For optimize
$$= yz + x(y \oplus z)$$

circuit :




# half substractor:

Truth Table

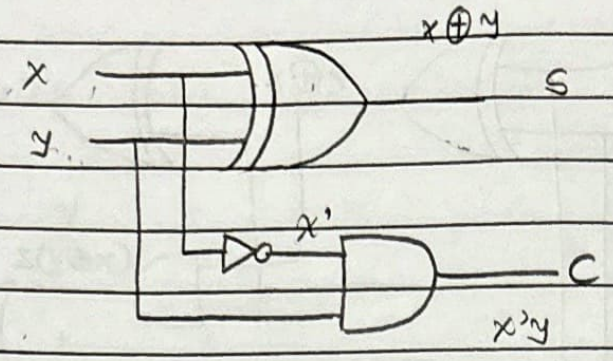
| x | y | S | B |
|---|---|---|---|
| 0 | 1 | 1 | 1 |
| 0 | 0 | 0 | 0 |
| 1 | 1 | 0 | 0 |
| 1 | 0 | 1 | 0 |

Expression :

$S = x \oplus y$

$C = x'y$

circuit :



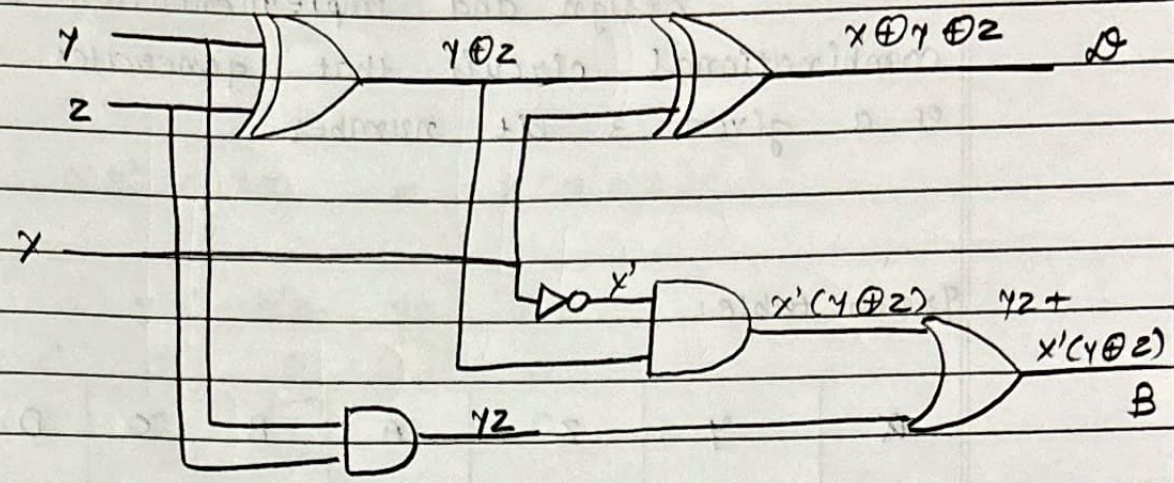

# full substracter:

Truth table

| X | Y | Z | D | B |
|---|---|---|---|---|
| 0 | 0 | 1 | 1 | 1 |
| 0 | 0 | 0 | 0 | 0 |
| 0 | 1 | 1 | 0 | 1 |
| 0 | 1 | 0 | 1 | 1 |
| 1 | 0 | 1 | 0 | 0 |
| 1 | 0 | 0 | 1 | 0 |
| 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 0 | 0 | 0 |

expression:

$$D = x'y'z + x'yz' + xy'z' + xyz$$
$$= x \oplus y \oplus z$$

$$B = x'y'z + x'yz + x'yz' + xyz$$
$$= yz(x'+x) + x'(y \oplus z)$$
$$= yz + x'(y \oplus z)$$

circuit :